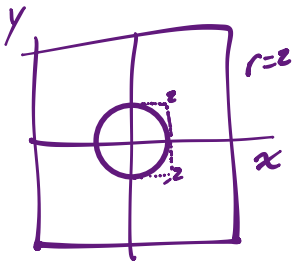
Starting with a circle.

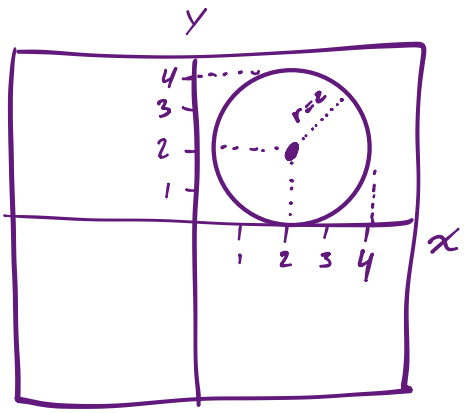
$x^2 + y^2$ = a circle of r^2 centered @ the origin. → (0,0)?

radius 2

$x^2 + y^2 = 2$ (centered @ (0,0))



$(x-2)^2 + (y+2)^2 = 2$ gives:



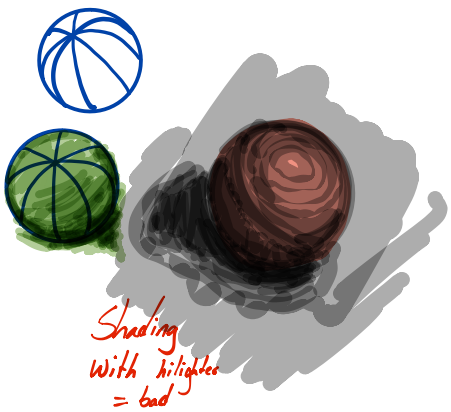
Now to go 3d just add z^2

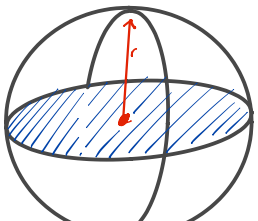
$x^2 + y^2 + z^2 = r$

radius = line segment from center to any point along edge.

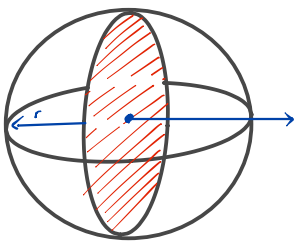
3d sphere centered @ (0,0,0)

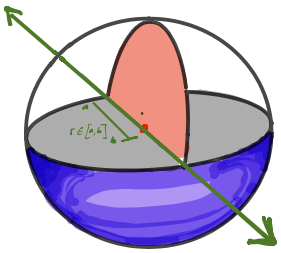
So if anything ever $= r^2$ then we know it's on the spheres surface.





$x^2 + y^2 + z^2$





Visually it now should be obvious why these 3 conditions hold.

Outside Sphere: $x^2 + y^2 + z^2 < r^2$

Inside Sphere: $x^2 + y^2 + z^2 > r^2$

On Surface Sphere: $x^2 + y^2 + z^2 = r^2$

Now to easily control where the sphere is at any point (C_x, C_y, C_z)

→ $((x+C_x)^2 + (y+C_y)^2 + (z+C_z)^2)$

Then lets make (x, y, z) & (C_x, C_y, C_z) all be vectors to more easily control.

Let $P = (x, y, z)$

$C = (c_x, c_y, c_z)$ — where we want to move to. Set the center of the sphere.

Now using dot we can do $(P-C)\cdot(P-C) = \|P-C\|^2$

L_2 norm, distance from P to C

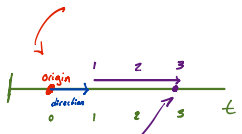
Every point P that satisfies $(P-C)\cdot(P-C) = r^2$ is on the surface of the sphere.

etc we can do inside if $> r^2$ and outside if $< r^2$

Rays. A start + vector

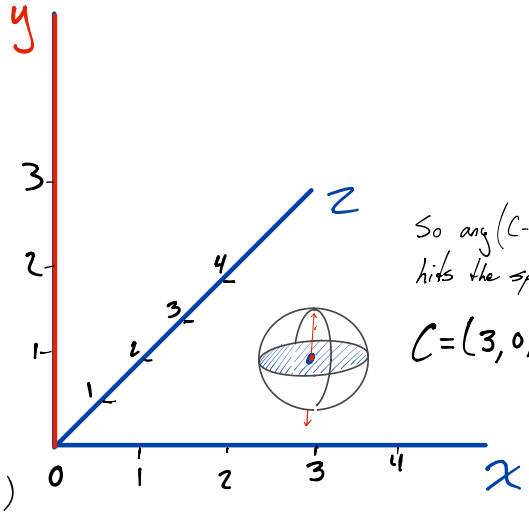
$P(t) = r_{ray_origin} + t\,d_{direction}$

$\hookrightarrow P(t) = r + tb$



$P(3) =$

This function places us (t·b) lengths along $(r+b)$



So $\text{any}\left((C-P(t))\cdot(C-P(t))\right) = r^2$ hits the sphere.

$C = (3, 0, 2)$

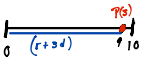
$r = (1,1,1) \quad d = (0,0,3)$

$P(0) = (1,1,1) + 0d = (1,1,1)$

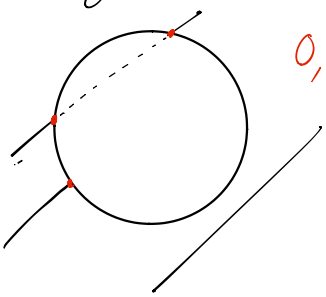
$P(3) = (1,1,1) + 3d$

$= (1,1,1) + (0,0,9)$

$= (1,1,10)$



Using the quadratic formula we can find how many intercepts there are one each ray



Summary.

$P(x,y,z)$ = hit point

$C(x,y,z)$ = location

$P(t)$ = some point along a vector

$\left((C-P(t))\cdot(C-P(t))\right) = r^2$ when $P(t)$ hits the sphere.

When doing $t = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

we will find the descrim will tell us either

$b^2-4ac < 0$ Shot ray miss

$= 0$ one intercept

> 0 two intercepts

$a =$

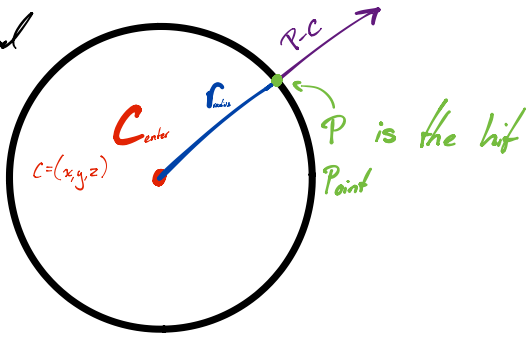
$b =$

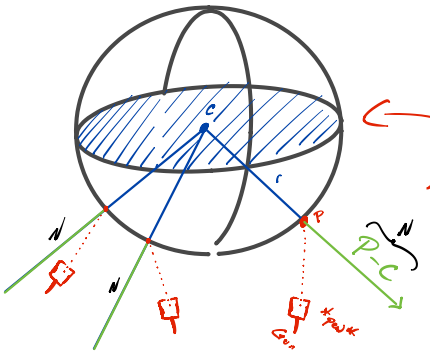
$c =$

Surface Normals

A unit vector perpindicular to the surface at the point of intersection.

Finding a normal for any hit point





lets shoot a ray and see the normal

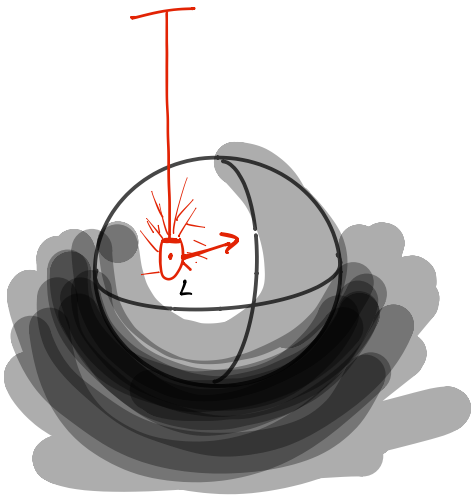
Let $\frac{P-C}{r} = N$

N for normal.

So we have units in a normalized form.

Now from $R_{ay}(t) = O_{rigin} + t\vec{d}$ (Point, vector)

We can do $\vec{N} \cdot \vec{d}$ to find the direction of the normal. Going in or out of the circle.



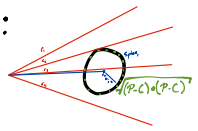
L is a vector for a ray from the light to the hit point.

Do $\max(0, \vec{L} \cdot \vec{d})$

if 1 it's hitting thus lit

Review

Now how to find if a ray intercepts with a sphere? → We know the location of the sphere, we know its radius, we know the ray vector → so find intercept:



My solution: Find length of vector from origin − Center of sphere. → Then that vector and the radius should be in the same world units. Wrong. World units is another thing.

Therefor we can... (change of thought)

New idea $(P-C)\cdot(P-C) = r^2$ (Point − Center of sphere), so if we know all values of $\sqrt{r^2}$

Then if $ray_{\alpha} = \sqrt{(P-C)\cdot(P-C)}$ we've hit the sphere? Almost, need to consider ray direction too.

> I dont understand what r^2 gives us. Radius, but now all values in a parabola? How to use r^2?

r^2 being a quadratic just tells us there is two roots.

Facts about r^2: $r^2 \in \mathbb{R}, \Rightarrow r \in \mathbb{R}$

if $x^2+y^2+z^2 = r$, then $x,y,z \in \mathbb{R}$.

~~if we set $r=2$ then let $G = \{p=2 \mid p = x^2+y^2+z^2\}$~~

~~Then G is the set of all combinations $x^2+y^2+z^2$ that $= 2$~~

$\Rightarrow$ ~~$x+y+z = r$~~ No

$\Rightarrow$ $x^3+y^3+z^3 = r^3$?

Let $A = \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 2^2\}$

or

$A = \{\vec{P} \in \mathbb{R}^3 \mid (P-C)\cdot(P-C) = 4\}$

Recall if $\vec{v}=(a,b)$ then $|\vec{v}| = \sqrt{a^2+b^2}$ then $|\vec{v}|$ is "normalized."

Sphere $S = \{\vec{P} \in \mathbb{R}^3 \mid (\vec{P}-\vec{C})\cdot(\vec{P}-\vec{C}) = 2^2\}$

Ray $R = \{\vec{P} \in \mathbb{R}^3 \mid \vec{P}(t) = \vec{O} + t\vec{d},\ t \geq 0\}$

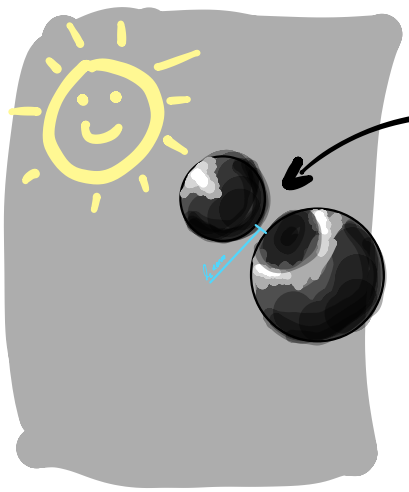
($\vec{P}(t)$: Vector notation on function? $t \geq 0$: So we don't go behind Camera.)

Need to find $S \cap R$

The points where both the sphere and Ray exist.

Ray w/ direction. $D = \{\vec{P} \in \mathbb{R}^3 \mid ((\vec{O}+t\vec{d})-C)\cdot((\vec{O}+t\vec{d})-C)) = 2^2\}$

$S \cap D$ = The intersection of rays on spheres.



Local Lighting

Shadow cast on one sphere by another

↳ When a normal is calculated, follow it and see if there are intercepts between it and a light source.

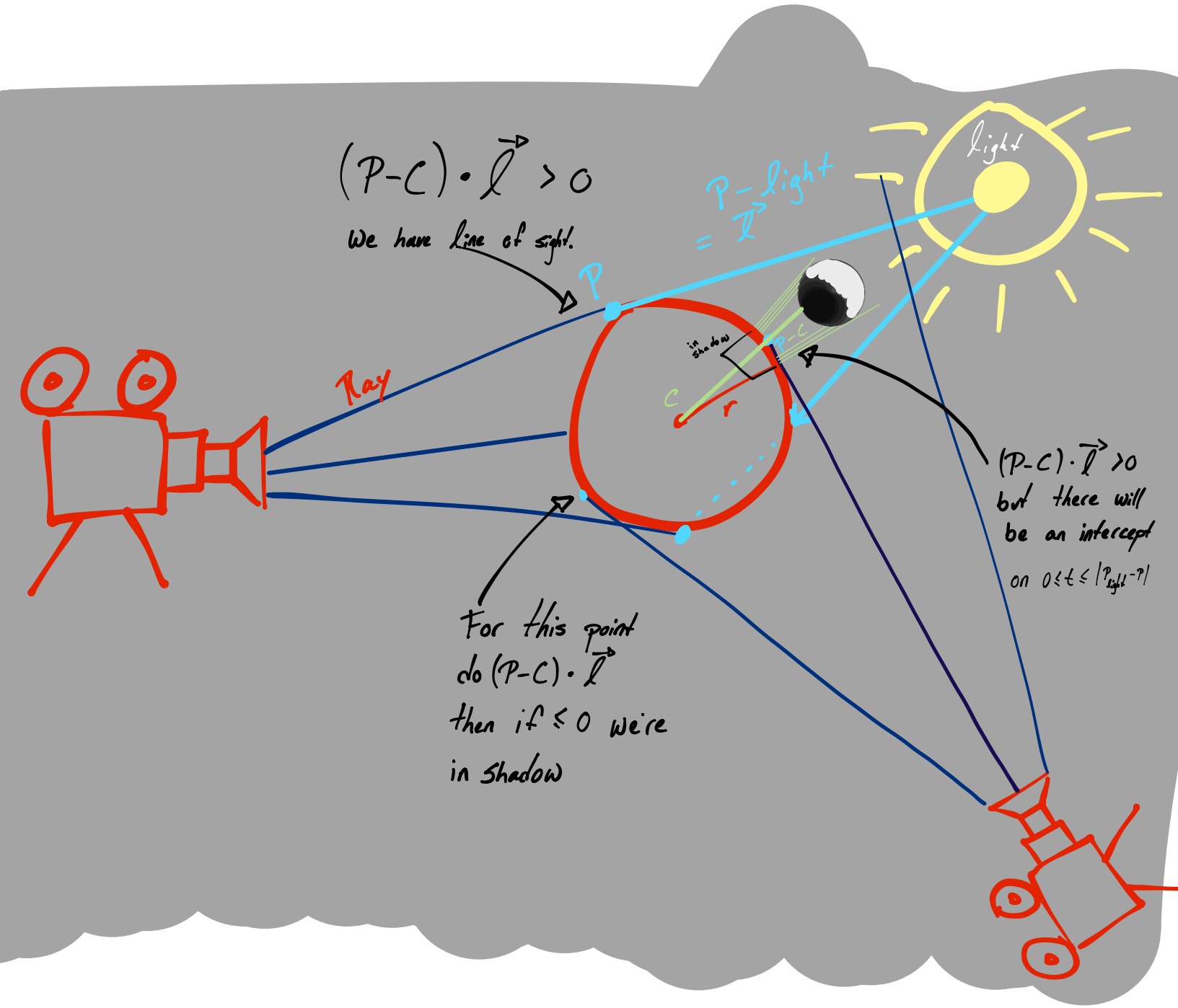
To calculate how much shadow look at the l_2 norm distance from the normal to the intercept.

Instead follow from hit point to light normalized: $\vec{l} = \frac{\vec{P}_{light} - \vec{P}}{|\vec{P}_{light} - \vec{P}|}$

Now we have $\vec{l}_n$ all the light rays from the camera rays.

So now we have all the $\vec{l}$'s we check for an intercept on each one to see if the hit point is in shadow

For rudementary brightness do $(P-C)\cdot\vec{l}$



Antialiasing

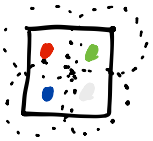
A pixel is not a little square!



This is often how we think of pixels, but in reality we see in $\mathbb{R}$ not $\mathbb{Z}$

Cameras resolution is limited so we capture eventually in $\mathbb{Z}$

In reality a pixel is a combination of area in $\mathbb{R}$



Ok... but how do we anti-alias....?

Right now I am doing whats called "Point sampling".

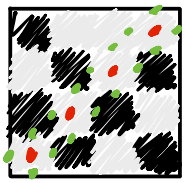
Shooting a single ray and setting a pixel for that first intercept.

The problem is this. How many rays to shoot.



Shoot 4 rays

We only see white



Instead shoot 4 rays for each one, each additional ray in a random direction by an offset of 0.5

Then average between them

